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MT. ST. HENRY'S PRINCIPAL AND SUBSIDIARY SEMINAR POWERED BY SUPER PLATINUM MATHEMATICS AND APEX EXAMNINATIONS COUNCIL

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PRINCIPAL MATHEMATICS 1
GEOMETRY

Item 1

A national athletics federation is conducting rigorous structural inspection on a newly completed multipurpose stadium. The main field is elliptical with a rectangular central field connected to two semi-circular tracks on its breadth sides.

2-Body
-Introduction
-Background

The bottom line of the longest side of the field passes through the points P(0, -50)m and Q(-40, -20)m while the top line passes through R(1, 44)m and S(5, 41)m. The inspection committee will approve this structure if the distance between these lines is at least 75m.

The field is parametrically modelled by:

$x = 5 + 100cos\theta$ and $y = 80sin\theta - 10$

where x and y are in terms of meters.

The field will pass the standards if its eccentricity is less than 0.7 to enable athletes negotiate the corners well and the distance between either focus and the centre of the ellipse is more than 55 m to enable long range javelin throws.

When an athlete is sprinting along the curved section of the track at position A, two automated camera tracking hubs are positioned at stations X and Y which are 160 meters apart to monitor the athlete's real time of performance.

The inspection committee will use the position of A when the athlete is exactly 100 m from station X and angle AXY = 60°. If the angle AYX is less than 40° and the distance from A to station Y is less than 200 m then the camera positions will be approved.

Task:

Help the inspection committee to make a mathematical report on the structure of the stadium.

Item 2

A final year civil engineering student who has been placed in the ministry of works for his industrial training.

The ministry has a project of constructing an express road to connect the three towns whose centre locations are on coordinates A(12, 13, 10), B(18, 20, 32) and C(20, 50, 50) with all coordinates being in miles.

He expects to construct a triangular network of a road to connect the three towns but he does not know whether the chosen centres will enable him achieve his goal. Furthermore, the plane containing these roads should be less than 0.5 miles from a trading centre whose centre coordinates are T(10, 15, 5) miles such that vehicles from this centre can easily access the road.

Additionally, the engineer wants to install traffic lights at points where the road network connects to the other roads in three major towns.

The system is programmed to switch the traffic lights from green to yellow the moment the signal strength is exactly 1 unit. The signal strength is

$S(t) = cos\left(\frac{\pi}{45}t\right) - \sqrt{3}sin\left(\frac{\pi}{45}t\right)$ for $0 \leq t \leq 90$

where t is the period for green light and yellow light to switch on with yellow light having less period in seconds than green light to prevent overstaying at the traffic light stations.

This system will effectively control traffic flow if the difference between the green and yellow light appearance is at most 30 seconds.

The project also includes designing a surveillance CCTV camera and the civil engineer is expected to identify the best location of this camera system along his triangular network.

He has spotted two points at locations R(3, 4) and S(5, 8) with all coordinates being in meters and he thinks that if the camera is positioned at point P(x,y) such that it is equidistant from the spotted points S and R, then the system will be able to even focus at a target point on location T(0, 8).

Task:

Help the engineer to make valid decisions on this project using your skills of geometry.

Item 3

Uganda electricity distributor company is carrying out an upgrade in the distribution of power in Kawempe. This is aimed at improving on their services to climate load shading and also add more customers to their grid.

There are already three existing service poles A, B and C located at (3, 10, 3), (5, 8, 4) and (3, 10, 5) respectively with all coordinates being in kilometres from the transformer. They need to add another pole for new customers at a point D(6, 5, 3) whose distance from the plane ABC should not exceed 2 km, so as to use the same transformer.

In another upgrade there is an existing linear network of electric wires running on a highway modelled by $y = 8 - 2x$ where x and y are in metres. They need to connect two other linear wires from a town located at E(6, -4) and both lines should make an angle of 45° with the existing line. One of these lines must connect new customers located at F(10, 8).

The existing transformer needs to be charged and replaced by a bigger one to cater for the new load and reduce on unnecessary load shading.

The new transformer to be installed has a thermostat which switches off power due to heating and switches off power immediately whenever the acute angle θ goes beyond 50°, it's modelled by

$$2cot^2\theta + 8 = 7cosec\theta$$

The transformer will only be approved if it doesn't cause any load shading by switching off itself automatically.

Task:

Help the company to know whether the required upgrades will be achieved.

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Item 4

A telecom company is expanding its infrastructure in your community. To ensure a successful rollout, you must evaluate the technical feasibility and regulatory compliance in three interconnected projects.

To maintain network stability, the company policy dictates a dynamic load-balancing intervention. If the peak active subscriber count on a single sector exceeds 20,000 concurrent users, management must trigger the "Off-Peak Data Bonus" campaign. This campaign provides data incentives exclusively between 12:00 AM and 5:00 AM to encourage heavy-bandwidth users to shift their usage away from peak operational hours and if the peak hour lies in this range, then the data bonus shifted to another time interval.

The total number of active subscribers (in thousands) over a standard 24-hour cycle is modeled by the wave function S(t), where t represents the number of hours past midnight ($0 \leq t \leq 24$):

$$S(t) = 13 + 6sin\frac{\pi t}{12} - 6cos\frac{\pi t}{12}$$

The telecom mast is to be installed at the centre of the path of a telecom wave which moves such that its distance from (2,0) is 3 times its distance from (0,4) with all coordinates being in kilometres. There is an existing mast at (20,2) and the company's policy before installation of new masts requires that the distance from the existing mast to the new mast must be greater than 20 km.

There is an existing road modelled as a straight line:

$$\frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+1}{-2}$$

With x, y and z being in kilometres.

The company has identified a location $A(2,1,-5)$ where it intends to put a support centre for its subscribers on condition that the shortest distance from the road to A is not more than 1 km to ensure easy access for walk-in customers.

Task:

Using your knowledge of geometry, help the telecom company to make valid decisions on their planned expansions.

Item 5

A certain radio telecommunication company is upgrading local radio cell phone towers to make the signal stronger and stop drop calls. Clever engineers are setting up a flat radio antenna panel and the position of this

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panel is marked by three points on a grid A(1, -2, 4), B(3, 1, 6) and C(2, 0, 7) all measured in meters. To avoid blocking signal this flat panel must be perfectly parallel to a straight radio communication beam modeled by:

$$r = \begin{pmatrix} 15 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

To protect workers from high energy radiations, the radio beam must be at least 7 m away from the panel.

Additionally, the engineers are mapping out a boundary zone between two neighboring radio signal towers located at P(1, 2)m and Q(6, -3)m. To balance the network load, a mobile signal testing unit must travel along a specific path such that the ratio of its distance from P to its distance from Q is 2:3. If the mapped zone contains the point K(3, 0)m, then the boundary zone will successfully separate the radio signals.

Furthermore, a specialized satellite dish at a main station uses a parabolic curved surface to focus incoming radio waves directly into a receiver point. The dish is modeled by:

$$y^2 - 8x - 4y + 2 = 0$$

where x and y are measured in hundreds of meters. The receiver must be placed exactly at the focus of the dish to capture the strongest signal. From this point, the dish links directly to a receiver antenna of the furthest radio located at R(16, 7). If the distance from the focus to this radio is less than 1500m, then the long signals will suffer from heavy environmental interference and won't be clear.

Task:

Help the engineers to make valid decisions on their upgrades.

Item 6

An electricity distribution company is carrying out an upgrade in their power distribution network system to improve their services and reduce local load shedding frequencies. The position vectors of three existing mainline transmission poles A, B and C are given respectively as:

$$\boldsymbol{OA} = -6\boldsymbol{i} + 7\boldsymbol{j} + 1\boldsymbol{k}$$

$$\boldsymbol{OB} = 2\boldsymbol{i} + 5\boldsymbol{j} + 3\boldsymbol{k}$$

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$OC = -3i + 4j + 5k$

The field engineers need to install another auxiliary distribution pole D whose coordinates must be chosen such all four poles lie on the exact same planar structural grid so that they can successfully interconnect to a single power supply transformer station. The proposed coordinates of D are (2, 4, 5).

Additionally, engineers are monitoring a major straight trunk line segment of the power moving grid moving directly along a highway modeled by:

$$2x + y - 8 = 0$$

They plan to install a new straight secondary line pathway from the main regional hub station located at a town E(3, 1) to connect directly to a regional utility maintenance centre F(5, 7). To proceed safely with construction, the engineers will require the angle between the two power lines not to be more than 50°.

The existing transformer gives fluctuating voltage, V with time, t in seconds is modeled by:

$$V = (240cos\frac{\pi}{10}t - 70sin\frac{\pi}{10}t)KV$$

This transformer is known for causing load shedding whenever its voltage reaches an absolute minimum and absolute maximum. The engineers will consider its replacement if the time interval between minimum absolute and maximum absolute voltages is less than 12 hours.

Task

Using your knowledge of mathematics, help engineers make a decision.

ALGEBRA

Item 7

Apex Global Solutions is auditing its regional head office operations to prepare for an upcoming corporate board review. The company's executive compensation matrix balances basic salary with gratuity and pay as you earn tax. The total amount of money paid to any member of the executive in t years is:

$P(t) = (4t^4 + 20t^3 + 28t^2 + 20t + 60)$ Millions

The basic monthly pay scaling factor is modeled by:

$B(t) = (2t^3 + 5t^2 + 10)$ millions

The company needs to know the corresponding pay as you earn tax function T(t) and the gratuity function G(t). The system modeled their computation as:

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$$P(t) = B(t)T(t) + G(t)$$

The company is designing a 3-year renewable contract and wants to know how much money the executive will pay in tax and how much gratuity the executive will get after 3 years.

The company will consider approving this payment plan if after 3 years both the tax and gratuity values are less than the cumulative of basic pay, the total tax is less than 15 million and the total gratuity is greater than 30 million. The company also plans to expand the office area of its executive. The architectural design of the office space gives floor area in tens of square meters as:

$A = \frac{256}{\sqrt{1-0.5x}}$ square metres

Where each term of the expansion represents floor area of each room and x is the reduction factor of the floor area of the rooms. The company needs only four biggest rooms with a reduction factor of 0.8, but this design will only be approved if the total area of the four required rooms exceeds 3000 square meters. The company has two regional branches $Z_1 = 3 + 4i$ and $Z_2 = 1 - 2i$ with all coordinates in tens of kilometers.

The company wishes to open another branch C at:

$$Z_3 = \frac{Z_1}{Z_2}$$

This branch will be cleared for operation if it satisfies $|Z_3 - Z_0| \geq 45\ km$

Where $Z_0 = -5 + 5i$ is the company headquarter.

Task:

Help the company to make a valid audit report using your algebraic skills

Item 8

A large-scale farmer always carries out intensive research before making any sowing decisions. He plans to plant maize on a large scale and the variety on market is modeled by:

$$P(x) = ax^3 + bx^2 + cx + 5$$

Where P is the total expected yield in thousands of kilograms and x is the mass in hundreds of the seeds sowed. Records from a farmer using the variety show that:

P(1) = 14, P(2) = 53 and (P(3) =146.

He plans to sow 600kg of seeds of this maize variety and his expectation is at least 900,000kg in yields. If this variety doesn't meet his expectations, then he will think of another variety.

The next season is expected to have limited amounts of rainfall and therefore he will need to use irrigation as an option and he would need a pump to be used for the irrigation. The pump on market pumps water at a reducing rate defined by:

$$V = \sqrt{\frac{(1+2x)}{(1-2x)}}$$

Where each term of the expansion represents volume of water pumped hourly and V is the amount of water in thousands of litres pumped at an hourly reduction rate of 0.8.

He plans to carry out his irrigation for only 4 hours daily and therefore he will consider using the pump if it can provide at least 8000litres of water for irrigation. The spray pipes used

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during irrigation give high pressure and cover a wider area depending on their cross-section area. He plans to use two pipes P and Q whose lengths are:

$L_p = \sqrt{4a + 5}$ and $L_q = 2\sqrt{(a + 4)}$

With pipe P being exactly 1 metre shorter than pipe Q during irrigation. The required pressure of water for irrigation can be realized only if the surface area of these pipes doesn't exceed 6 cm^2. If these pipes don't meet the required measurement the farmer will think of other means of increasing the water pressure.

Task:

Using your knowledge on algebra help the farmer make valid decisions on his quality checks.

Item 9

A farmer planted large amounts of coffee on a large estate and after three seasons he is carrying out an audit on the farm to make some important decisions. The coffee variety that he planted gives yields depending on the amount of fertilizer applied per acre. The total yield (y) in thousands of kilograms after applying hundreds of kilograms of fertilizer per season is modeled by:

$$y = ax^2 + bx + c$$

In the first season he applied 200kg of fertilizer and he yielded a total of 193,000kg of coffee, while in the second season he applied 400kg of fertilizer and total yield was 333,000kg of coffee while in the third season he applied 500kg of fertilizer and yielded 388,000kg of coffee. He will consider uprooting this coffee variety if the expected maximum yield does not exceed 500,000kg of coffee.

He also uses a machine to pump irrigation water from the storage tank into the coffee plantations. The pipes used for irrigation have equal length with:

$L_A = 3^{(r^2+9)}$meters and $L_B = 729^r$ meters

Where r is the radius of the pipes in cm.

These pipes can only spray a wide area with enough pressure if their radius not exceeding 4cm, otherwise he will consider replacing them.

The water accumulates in the underground tank during the night and during day it's pumped into the plantations in such a way that the amount of water pumped per hour reduces by constant ratio every hour. During the first hour 5,000 liters of water is pumped into the plantation and this amount reduces to 2,560 liters in the fourth hour. He irrigates his coffee plantations for 8 hours every day and if the total amount of water pumped is less than 19,000 liters he will consider another source of water.

Task:

Using your knowledge of algebra help the farmer to make a valid audit report.

Item 10

A water engineer is making designs for a new water pump system. A water engineer has been contracted to dig a water well and then install a system to pump the water into a tank

mounted on a water tank stand. The formula he uses to reach water levels is quadratically modelled by:

$$V = ah^2 + bh + c$$

Where V is the expected volume in hundreds of litres obtained at a depth h in hundreds of meters underground. He uses the formula to dig wells for three customers at different heights.

The records of the water obtained and the depth of their water wells are recorded in the table below:

customer	Depth of the well × 100m	Volume of water obtained× 100 litres
A	1	90
B	2	168
C	3	234

The new customer needs a well which can provide at least 35,000 litres of water for his large poultry farm. The water engineer will only use this formula for this project if the maximum amount of water it can provide can meet the customer's demands.

Additionally, the water engineer is to install pipes for the customer.

The pipes he plans to use are of lengths $L_1 = log_4^r$, $L_2 = log_2^{2r}$ and $L_3 = log_8^{16r}$ whose total length is 6 meters, and r is the radius of the pipes in cm. These pipes can only provide the required pressure of water for the customer if their radius does not exceed 5cm; otherwise, the engineer will consider another set of pipes.

The customer has a section of poultry farm and the engineer wishes to install 8 different types of specialized water filters. To protect poultry from many diseases, the engineer needs to select 3 distinct filters to build a multi-stage filtration unit. He can only meet the customer's needs if the number of possible selections yields more than 50 selections.

To power the massive water pump, the engineer will install an alternating current (AC) motor.

The pump motor has resistance $R_1 = (6 + 2i)$ ohms.

The capacitor has resistance $R_2 = (2 - 4i)$ ohms. For the motor to operate safely, the ratio of the pump motor resistance to the capacitor resistance $\frac{R_1}{R_2}$ should give a real part less than 0.5 and the modulus of this ratio should not exceed 1.5.

Task:

Help the water engineer to make valid decisions on his new project.

Item 11

Alex is a large-scale poultry entrepreneur who uses an artificial intelligence-powered feed management system to forecast the duration of feed stocks based on historical consumption data. Agricultural specialists established that the duration, $P(x)$, in days, for which a stock of x tonnes of feed last can be modelled by

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$P(x) = x^4 + ax^2 + bx$

where a and b are constants. Farm records indicate that 2 tonnes of feed sustained the birds for 30 days, while 4 tonnes of feed sustained the birds for 300 days. The manager plans to purchase 5 tonnes of feed for the next production cycle. However, he is not certain whether these feeds can be enough for at least 600 days.

Alex also launched a pilot programme using an organic enzyme substrate to grow liquid bio-fertilizer cultures in a laboratory setting using the poultry droppings. During the first three days of monitored incubation, the daily yield measured 20 litres, 24.5 litres, and 30.0125 litres, respectively. Biochemical extension officers confirm that under optimal feeding protocols, this bacterial strain follows a strict geometric progression. He needs not less than 7000 litres of this fertilizer in 21 days to be supplied to coffee farmers but he doesn't know whether this will be possible.

On his farm, he has 7 male and 5 female staff who work daily and he wishes to make a rota of 9 staff daily so that they can have day and night shifts and some days off. This rota will only be possible if in any shift the number of male staff is more than that of female with atleast 3 female per shift. This arrangement is to work on a monthly basis and it will only be effective if the possible number of shifts exceeds 160.

Task: Based on algebraic reasoning, advise Alex on the decisions to undertake.

CALCULUS

Item 12

Emma is a graduate of agriculture who, after failing to get a job in his profession, decided to start tomato farming. He plans to use a customized watering can for watering his tomato plants modeled by:

$$4y = (x-3)^2$$

where x is the horizontal radius in cm of the can and y is the depth in cm. The can has a flat base and a maximum height of 30cm. Emma will consider using this watering can if its volume is not less than 10 litres.

Two days back, he discovered that a strange viral blight had affected 20 out of his 500 tomato plants, and today he has discovered that 30 more plants are affected. This virus is said to spread at a rate proportional to the square of the plants remaining uninfected. He plans to spray the garden in a few days to come, which will make it exactly 12 days since he discovered the viral infection. His spraying schedule will be considered late if, by then, more than 40% of the plants are affected.

Emma plans to secure an agricultural expansion loan, but he must analyze his market risk factor (R) that accumulates over the weeks (w) he delays his sales. This is modeled by:

$$\frac{dR}{dw} = \frac{320}{(12+w)(20-w)} \quad for\ 0 \le w \le 10$$

where he will only secure the loan if the maximum risk factor is less than 15.

Task:
Help Emma to make valid decisions on his farm using calculus.

Item 13

A fresh graduate civil engineering graduate has been offered a contract to supply bricks for a 3-storey building. He is expected to obtain the stone dust for making his bricks by excavating a rock modelled by $y = 4 - 25x^2$

Where x is the horizontal distance from the central axis of symmetry and y is the perpendicular height of the excavation channel from the flat of ground level, both measured in tens of meters. The engineer expects to make bricks using stone dust of volume not less than $1000m^3$. If the rock doesn't provide enough stone dust, then he will consider other options for top up.

The engineer plans to use a concrete brick making machine whose number of bricks N in hundreds it produces in t hours is related by:

$$N = (4-t)^2(1+t)^3$$

The machine does not operate for long hours to prevent overheating. This will help the engineer to make the target number of bricks as soon as possible on daily stop operation. It can produce at least 10,000 bricks. The engineer also plans to use a crane for loading the baked dry bricks onto the delivery trucks.

The crane loads the bricks at a rate:

$\frac{dN}{dt} = \frac{t^2}{t^2+1}$ $for\ 0 \le t \le 10$here N is the number of bricks in tens of thousands loaded after t hours of continuous loading. The engineer's expectation is to load at least 8,000 bricks per hour.

Task:
Using your knowledge of calculus, help the engineer know whether he will achieve the goals.

Item 14

An ICT student was tasked to set up a local high-speed streaming media server for upcoming school pavilion festival. The server needs to stream movies and music content across five distinct playback quality modes to accommodate different students' devices whose sizes are in descending order; 4K ultra-HD (A), 1080p HD (B), 720p HD (C), 480p standard (D) and Audio only mode (E).

To handle multiple users gracefully, the data bandwidth for these streaming corresponds to the first five terms of the expansion of:

$$C(x) = 155(4e^x - 9\ln(1-x))$$

where C(x) is measured in Mbps and x represents the network optimization index and each term of the expansion represents amount of data consumed by each of the above devices in the given order.

The ICT laboratory has allocated 3000 Mbps to this ICT student but to prevent the entire network from lagging, the ICT student plans to allocate at most 80% of the Mbps allocated.

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treaming. If the devices to be connected go beyond that percentage, then he will consider reducing them to fit in his plans when x = 0.6.

ensure full internet access across the pavilion layout, the expansion speed of Fi radius (r) over the set wave angle interval is modeled using:

$$\frac{dr}{d\theta} = \frac{80}{5 + 3cos\theta} \; for \; 0 < \theta < \frac{3\pi}{4}$$

gnal radius in meters and is θ the wave angle phase in radians. This set Wi-ure full internet access if its radius is at least 30m.

lent wants to make sure the streaming app does not slow down his laptop k. The app features an automatic garbage collection program that actively porary storage junk. The remaining junk at the end of every movie is tracked on:

$$\frac{dM}{dt} = \frac{-6Mt}{(t^2 + 1)}$$

residual junk in MBs remaining in the student's laptop storage at any time t he first used it, for a similar purpose, the residual junk was 800MBs initially sider using it again for the festival if the residual junk reduces to less than ours.

idents make decisions using your knowledge of calculus.

: in agriculture has decided to set up a model farm on the family land in the n is to have five distinct gardens on which he will plant Tomatoes (T), Onions , green pepper (P) and Garlic (G).

gardens are designed to be in the strict descending order:

$$T > O > C > P > G$$

s designed the sizes of these gardens basing on his land to correspond to the ansion of the area function:

$)^4 + (1 + 3x)^4]$ square meters

eduction percentage of the gardens.

it has land of total area measuring 3000 square meters and doesn't want to cultivation because part of the land must contain the farmhouse, water tank nd dedicated nursery beds for raising young seedlings. He will therefore plan if the biggest four cultivated gardens cover an area not exceeding 70% using an 0.8 reduction rate.

or effective irrigation of his plants, he plans to install a central Irrigation Spray ter engineer has proposed a calibrated spray model where the rate of change ius r (in meters) over time t (in hours) is given by the equation:

or $0 \leq t \leq 3$

plete and healthy crop growth, the radius of this circular spray should be at in order to cover all the five gardens effectively.

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Furthermore, the agriculturist expects to mix water-soluble fertilizers in the water before is sprayed during irrigation. The amount of fertilizer M(in Kilograms) in the tank at any time t (in hours) is measured by:

$$\frac{dM}{dt} = 5 - \frac{19M}{4}$$

He expects to dissolve 15kg of the fertilizer in the water daily. By the end of an irrigation time of 3hours, if the amount of fertilizer remaining in the water is less than 1.5kg, he will be sure that the plants have received enough fertilizer nutrients.

Task:

Using your knowledge on calculus, help the Agriculturalist to make valid decisions on his model farm plans.

Item 16

A group of tourists in a certain country were trekking to go up a 1000 m mountain. Their tour guide studying the safety briefing told them that at the mountain's foot the temperature is 16°C, and that the rate at which temperature drops with height on this mountain is modelled by the equation:

$$\frac{d\theta}{dh} = kh$$

The tour guide further informed the team that the temperature at the peak is -9°C.

During the briefing, one of the tourists wished they could go to at least 900 m up the mountain but the tour guide told them it's not safe to go beyond the height at which water starts to freeze. The tourists were wondering at what height does water start freezing.

One of the tourists had her water in a transparent bottle of height 30 cm and the volume of water in the bottle was defined by:

$$V = \pi\left[\frac{(h + 30)^3}{300} - 90\right] cm^3$$

where h is the depth of the water remaining in the bottle.

When the bottle was halfway filled, she realized that she was drinking at a constant rate of $-10\pi \; cm^3 min^{-1}$ and she was to continue taking water from the same bottle in the next 20 minutes before changing to another bottle but this was only to be possible if after this time the height of the water in the bottle was not less than 1cm otherwise the remaining water bottles would not be enough for the journey.

During this trip to the mountain, Carol a tourist on board had wanted to buy a circular necklace from one of the crafts shops. The inner radius of the necklace was 9.8cm as opposed to her neck radius of 10cm. Carol believes that if the error in the area of this necklace doesn't exceed $0.5cm^2$, she can buy the necklace for it will fit her.

The area of the necklace is modelled by

$$A = ln\sqrt{\frac{1 + 2x}{1 - 2x}}$$

Where x is the radius of the necklace.

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Task:

a) Determine whether the tourists reached 900m up the mountain or not.
b) Did the tourist have enough water for the journey.
c) Determine whether, Carol bought the necklace or not.

PRINCIPAL MATHEMATICS 2

SECTION A
ITEM 1

A construction company is overseeing the development of two access roads within a newly planned residential estate in Iganga. For planning and surveying purposes, the roads are modelled by the functions

$$y = e^x \text{ and } y = 3 - x, \text{ for } 0 \leq x \leq 1$$

Where the horizontal distance x is in kilometres.

The engineering team intends to install a traffic signpost at the exact point where the two intersect along the x-axis. Determining this junction point is important for road navigation, traffic control, and placement of street infrastructure within the estate.

Furthermore, the land enclosed between the two road profiles over the interval $0 \leq x \leq 1$ is scheduled for compulsory acquisition to allow construction of drainage channels, pedestrian walkways, and utility lines. To ensure transparency and fairness in compensating affected landowners, the company must accurately determine the enclosed area.

The required land area is divided into two sections:

- the area under $y = e^x$ from $x = 0$ to the junction point, and
- the area under $y = 3 - x$ from the junction point to $x = 1$.

For financial accountability, the surveying department will compare the exact area with a numerical approximation obtained using 4 subintervals, correct to three decimal places. The compensation process will be considered fair if the percentage error between the approximate area and the exact area is less than 1%.

Tasks

(a) Using a graphical approach, estimate the x-coordinate of the junction point where the signpost will be installed, correct to 1 decimal place.

(b) Using a suitable iterative method, determine the exact x-coordinate of the junction point, correct to 3 decimal places.

(c) Assess mathematically whether the compensation offered by the company is fair according to the stated criterion. **Item 2**

An estate's company recently purchased a rectangular plot of land for future investments. The plot owner says, his plot measures 104.65 m by 26.2 m, with both dimensions correct to the nearest decimal place.

The company's technical team is required to determine the possible range of the actual area of the plot, and to assess whether the absolute error in the estimated area does not exceed 5 m^2, in order to approve payment.

The company is also reviewing its annual financial performance to guide future investment decisions and tax planning.

The financial estimates are as follows:

- Estimated annual profit = 25 million UGX, with a margin of error of ±3%
- Estimated annual loss = 4 million UGX, with a margin of error of ±2%
- The Uganda Revenue Authority has assessed an annual income tax of 3.5 million UGX

The management wants to determine the possible percentage range of gross income that goes to tax.

To improve employee welfare and long-term financial security, the company introduces a staff savings scheme. Employees can invest a portion of their salary, which earns compound interest annually at a rate of r%.

The savings model is given by:

- Initial investment: P
- Annual interest rate: r%
- Duration: $n = 5$years
- Amount after $n$years: A_n

The company uses the compound interest model to track growth and encourage participation in the scheme.

To automate the savings scheme, the company requires a clear flowchart-style algorithm that:

1. Reads the initial amount P_1, previous amount A_1, and interest rate r
2. Iterates the compound interest process for n = 5 years
3. Computes and prints the accumulated amount A_n

The correctness of the algorithm is verified using a test case for a period of 5 years:

- $P_1 = 100{,}000$UGX
- $r = 15\%$

Tasks.

As a student of math perform a mathematical analysis to;

(a) Assess whether the plot will be bought

(b) Determine the percentage range of gross income that goes to tax

(c) Design a flow chart algorithm and verify its correctness.

SECTION B
PART I

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ITEM 3

The school administration of a certain a secondary school is reviewing the performance of candidates in the recently conducted resourceful examinations. The administration intends to use the results to guide decisions on remedial teaching, resource allocation, and academic interventions for the learners.

The marks obtained by the students have been summarized in the grouped frequency table below.

Marks	$0-<15$	$-<30$	$-<35$	$-<40$	$-<55$	$-<65$	$-<75$
Number of students	6	8	15	20	10	7	4

The administration requested the class teacher to provide a summary of the median and average mark, would wish the magnitude of their difference to be more than the standard deviation for the remedial lesson to be conducted.

When this data was entered in the computer, it produced an approximately normally distributed data, the administration wanted to know 80% confidence interval around the mean mark, and the probability between 80% confidence interval would be a probability of success for the sample of 15 selected students for which they wanted to know the probability of at least 13 students being successful and would wish probability to be less than 0.3, to recruit more math teachers.

TASK:

As a mathematics student provide the necessary information to the school administration to make informed decisions.

ITEM 4

In Kampala Northern By-pass, road accidents are increasing due to dangerous practices such as reckless crossing by pedestrians and over-speeding by motorists. Consequently, traffic-control measures, including traffic lights and road humps, have been introduced on busy roads.

At a particular point along the by-pass, the speeds of cars are normally distributed. Police investigations established that: 95% cars travelled at a speed less than 85 kmh^{-1} and 10% travelled at a speed less than 55 kmh^{-1}. According to traffic regulations, any car travelling at a speed exceeding 65 km h^{-1} shall be impounded.

Further, the traffic police sampled 100 cars along the by-pass to investigate their average speed. The Officer in Charge of Traffic directed that new road humps would be constructed if the proportion of the cars travelling at an average speed greater than 70 km h^{-1} exceeds 0.45.

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In addition, the police inspected 300 cars that were in dangerous mechanical condition (DMC). The ages of these cars, X, measured in tens of years, were found to follow the continuous probability distribution given by

$$f(x) = \begin{cases} k, & 0 \le x \le 1.5 \\ \frac{k}{2}(2-x), & 1.5 \le x \le 2 \\ 0, & \text{else where} \end{cases}$$

Cars older than 1.6 tens years will be taken to the police station for further forensic analysis.

Tasks

As a senior six student if consulted, come up with a clear report on the following backed by calculations:

(a) The percentage of cars that shall be impounded

(b) Whether new road humps will be constructed or not.

(c) The number of DMC cars that were taken for forensic analysis

Item 5

During a consumer survey, a research team is studying customer preferences for two different television service providers, **DStv** and **StarTimes**, in two different areas.

- **Urban area:** contains 4 respondents who use **DStv** and 5 respondents who use **StarTimes**.
- **Rural area:** contains 4 respondents who use **DStv** and 3 respondents who use **StarTimes**.

Due to differences in the number of subscribers, the **urban area is twice as likely to be selected as the rural area**.

A researcher randomly selects one area and then randomly interviews two subscribers from that area, with each subscriber being interviewed only once.

The research team wishes to make an assessment on:

- the probability that the two selected subscribers use the **same television service provider**;
- the probability that the two subscribers selected are from the **urban area**, given that they use the same television service provider;
- the **probability distribution** of the number of subscribers who use **StarTimes**, and the **expectation** of the distribution.

At the same time, a television service provider is conducting a survey in different urban areas to determine the relationship between the number of people who use **StarTimes** and the number of people who subscribe to **DStv**. The findings were summarized in the table below.

DS TV (x)	78	66	73	75	84	66	89	84	67	77
Star Times TV (y)	81	68	81	75	80	67	85	83	66	78

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The research team wishes to:

- represent the information graphically and draw a **line of best fit** to determine the relationship between the numbers of StarTimes and DStv subscribers;
- estimate the number of **DStv subscribers** if there are **80 StarTimes subscribers**;
- change their marketing strategy if the **rank correlation coefficient is less than 0.8**.

Tasks

As a mathematics student, help:

(a) The research team make the required probability assessment.

(b) The television service provider's marketing team perform the statistical analysis and determine whether they should maintain or change their marketing strategy.

SECTION B

PART II

ITEM 6

With the aim of ensuring that the transport truck arrives at its destination on time, the site engineer, plans the journey from Town A, B and C. A Sino truck carrying construction bricks passes through three town A, B and C which lie on a straight road with AB=8.5 km and AC=28 km. The truck moves along the straight road with a constant acceleration of $^1/_{80}\,ms^{-2}$, it also take it 10 minutes to travel from A to B. For the truck to make two trips of bricks, the time taken for the truck to move from town B to town C should be less than 30 minutes.

At construction site, bricks are raised up the building site by means of a vertical rope tied to one end of light bucket. At every trip, a pack of bricks of mass 100kg is to be hoisted up the side of a building at an acceleration of $0.25ms^{-2}$. The process of raising the bricks will be considered safe and free from danger if the tension in the string attached to the bucket is more than 1050N.

Additionally, after offloading the bricks, the bucket is returned to the ground by throwing it from the edge of the vertical building 52.5m high with a velocity of $2\ ms^{-1}$ at an angle $\tan^{-1}\frac{7}{24}$ above the horizontal. The bucket hits ground at a point from the base of the building. The site engineer wants to ascertain the safety of the site workers who are standing 10 m from the base of the building.

Task

Help the engineer to obtain the mathematical information required to make informed decisions.

ITEM 7

A **sports stadium management team** plans to install a rectangular metal **advertising board** on the roof of the stadium. An engineer is hired to assess the stability of the board before it is installed. The board has specified rectangular dimensions $AB = 0.5$ m and $AD = 1.5$ m.

To assess its structural stability, forces of magnitudes 15 N, 12 N, 16 N, 10 N, and 20 N act along the sides AB, CB, CD, AD and the diagonal AC of the board respectively. According to the stadium's safety guidelines, the resultant force acting on the board must be at least 20 N for it to remain securely fixed.

During a severe windstorm, the advertising board becomes detached from its mounting and begins to slide down a rough section of the stadium roof. The roof is inclined at a 20° to the horizontal and has a coefficient of friction of 0.3. Starting from rest, the board slides for 2 seconds before reaching the edge of the roof, which is at a height of 3 m above the ground.

The engineer observes that the board leaves the roof and travels through the air at an angle of 30° **below the horizontal**. A **flower garden** is located on the ground, with the nearest flowers planted **1 m from the stadium wall**. The stadium management is concerned that the falling advertising board may land on the flowers.

Task

As a **mathematics student**:

(a) Advise the engineer whether the advertising board is securely fixed based on the forces acting on it.

(b) Determine whether the advertising board will fall on the flowers.

ITEM 8

During a joint marine surveillance and rescue training exercise on Lake Victoria, a patrol team from the Uganda Police Marine Unit is monitoring the movement of vessels and coordinating an emergency cargo delivery operation.

A patrol boat is travelling at a speed of 75 km h^{-1} when its driver notices a transport aircraft flying horizontally at 180 km h^{-1} at an altitude of 900 m dropping emergency cargo supplies to a ship travelling at 20 km h^{-1} in the direction $N40°E$

At the instant the cargo is released, the patrol boat is located 5 km from the ship on a bearing $S30°E$ from the ship.

The patrol boat driver intends to intercept the ship before the cargo lands so as to provide security support and assist with the recovery of the emergency supplies.

When the cargo finally lands on the deck of the ship, it continues to slide along a rough horizontal surface before coming to rest. The coefficient of friction

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rgo container and the deck surface is 0.3 and the cargo slides
ıce of 10 m before stopping completely.

ıtics student assisting the marine surveillance team, you are

ıe the course that the patrol boat driver should set in order to
the moving ship.
ıe mathematically whether the patrol boat reaches the ship before
lands.
ıe the speed of the cargo immediately after landing on the ship's
the time for which the cargo slide

END

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